

Practical Activity: Shopping Bill Calculator

یہ شاپنگ بل پروگرام تینوں کنٹرول اسٹرکچرز کو ملا کر بنایا گیا ہے: while لوپ سے اشیاء بار بار شامل ہوتی ہیں، ...if

CS XI

Chapter 3: Programming Fundamentals

Lecture 20 of 24

Practical Activity: Shopping Bill Calculator

LEARN IT

This program combines all three control structures: **sequence** (steps in order), **selection** (checking "done", checking the price, checking for a discount) and **repetition** (a while loop that keeps adding items). It uses a **flag** — a Boolean variable (shopping) that controls when the loop stops. `item.lower()` converts the text to lowercase so that "done", "Done" and "DONE" all work.

```
print("==== SHOPPING BILL ====")
total = 0.0
count = 0
shopping = True

# REPETITION: keep adding items
while shopping:
    item = input("Enter Item Name (or 'done' to stop): ")
    # SELECTION: check if user wants to stop
    if item.lower() == "done":
        shopping = False
    else:
        price = float(input(f"Enter Price of {item}: "))
        # SELECTION: validate price
        if price <= 0:
            print("Invalid price!")
        else:
            total += price
            count += 1

# Summary and discount
if total >= 5000:
    discount = total * 0.20 # 20% off
else:
    discount = 0

final = total - discount
print(f"Total Items: {count}")
print(f"Total Bill: {total}")
print(f"Final to Pay: {final}")
```

WORKED EXAMPLE

Sample run. The user buys Rice for 3000, types a wrong price -50 for Soap, buys Oil for 2500, then types DONE :

OUTPUT

```
==== SHOPPING BILL ====
Enter Item Name (or 'done' to stop): Rice
Enter Price of Rice: 3000
Enter Item Name (or 'done' to stop): Soap
Enter Price of Soap: -50
Invalid price!
Enter Item Name (or 'done' to stop): Oil
Enter Price of Oil: 2500
Enter Item Name (or 'done' to stop): DONE
Total Items: 2
Total Bill: 5500.0
Final to Pay: 4400.0
```

Step	item	price	total	count
1	Rice	3000.0	3000.0	1
2	Soap	-50.0 (invalid, not added)	3000.0	1
3	Oil	2500.0	5500.0	2
4	DONE → shopping = False	—	5500.0	2

After the loop: $5500.0 \geq 5000$, so $\text{discount} = 5500.0 \times 0.20 = 1100.0$, and $\text{final} = 5500.0 - 1100.0 = \mathbf{4400.0}$.

REMEMBER IT

Label the parts of any program with comments, as here: # REPETITION , # SELECTION . It helps you, your teacher and the examiner see the structure.

COMMON MISTAKE

`total` and `count` must be set to 0 **before** the loop. If you put `total = 0.0` inside the loop, it resets on every item and the bill is wrong.

اردو میں سمجھیں

یہ شاپنگ بل پروگرام تینوں کنٹرول اسٹرکچرز کو ملا کر بنایا گیا ہے: while لوپ سے اشیاء بار بار شامل ہوتی ہیں، if-else سے غلط قیمت روکی جاتی ہے اور 5000 یا زیادہ کے بل پر 20 فیصد رعایت دی جاتی ہے۔

CHECK YOURSELF

1. What will "Final to Pay" be if the user buys one item for 4000 and then types done ?
2. What is the purpose of the variable shopping ?

Answers: 1. Final to Pay: 4000.0 (4000 is less than 5000, so no discount). 2. It is a flag that keeps the while loop running; setting it to False stops the loop.

Next lecture: 3.6 Libraries in Python